



Multi-scale RWKV with 2-dimensional temporal convolutional network for short-term photovoltaic power forecasting

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ABSTRACT

Improving the accuracy of Photovoltaic (PV) power forecasting is crucial for optimizing the schedule of power stations and maintaining the grid stability. However, PV power generation exhibits complex periodicity and is significantly influenced by weather conditions, introducing instability, intermittency, and randomness, making accurate PV power forecasting a challenging task. Therefore, this study proposes a multi-scale Receptance Weighted Key Value with 2-Dimensional Temporal Convolutional Network (MSRWKV-2DTCN) for PV power forecasting, which can learn periodicity and interdependencies of data and improve forecasting accuracy. Firstly, the proposed model identifies multi-periodicity of PV power data with the Fast Fourier Transform (FFT). Subsequently, we combine these identified periods with the canonical time mixing block of Receptance Weighted Key Value (RWKV) and introduce a multi-scale time mixing block to learn periodicity of data. Finally, to explore complex interdependencies of historical data, we replace the channel-mixing block of RWKV with a multi-scale 2-Dimensional Temporal Convolutional Network (2D TCN). Experiments were conducted on real-world datasets collected from Yulara solar system in Australia to validate the performance of the proposed model. Comparisons with other PV power forecasting models and ablation studies confirm that the MSRWKV-2DTCN achieves higher accuracy in short-term PV power forecasting.

1. Introduction

As global energy demands continue to rise, there is a growing shift towards alternative renewable energy sources aimed at reducing greenhouse gas emissions. Solar energy, being both renewable and environmentally friendly, is playing an increasingly crucial role in energy systems [1,2]. Solar power is an ideal solution for densely populated urban areas, serving as a pivotal component in achieving a net-zero carbon energy supply by 2050 [3]. Nevertheless, the output of PV power systems has complex periodicity and fluctuates with external meteorological factors, resulting in intermittency, instability and randomness, which presents significant challenges for the stability and balance of the grid [4]. Therefore, enhancing the precision of PV power forecasting is important for the effective dispatch and scheduling of the power grid. The framework for PV power forecasting is illustrated in Fig. 1.

According to the classification approach for PV power forecasting models suggested in many studies [5–7], we divide these models into three categories: physical models, data-driven models, and hybrid models. The classification is shown in Fig. 2.

Physical models, as known as “white box” approaches, leverage mathematical equations and algorithms to model the fundamental physical processes that convert solar energy into electrical energy within a PV system [8]. According to the input of physical models, it can be classified into three categories: satellite imaging, sky imagery, and Numerical Weather Prediction (NWP) [5]. Although physical methods can provide interpretable predictions, these models are particularly susceptible to abrupt weather fluctuations and lack flexibility, delivering unreliable predictions in the event of significant changes in weather conditions [9,10].

Data-driven models utilize historical PV output data and historical meteorological data to forecast the future output of PV systems. Data-driven models include persistence, statistical, and machine learning models. The persistence model makes a simple assumption that the current PV output determines the output of the next day. Statistical models utilize historical data and attempt to figure out mathematical patterns of the historical data. Popular statistical models include Markov chain [11], Autoregressive (AR) [12], Exponential Smoothing (ES) [13], Autoregressive Moving Average (ARMA) [14], fuzzy logic [15], and

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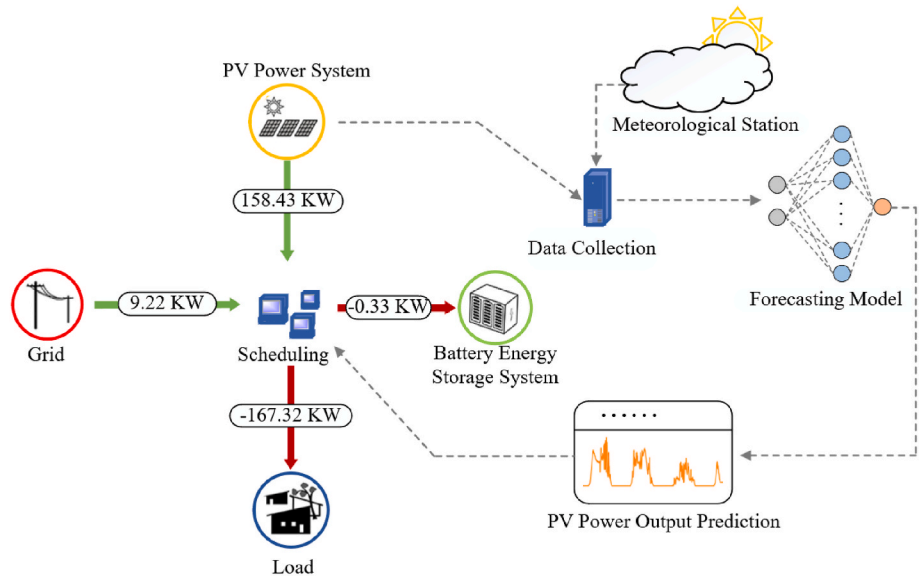


Fig. 1. Framework of PV power forecasting.

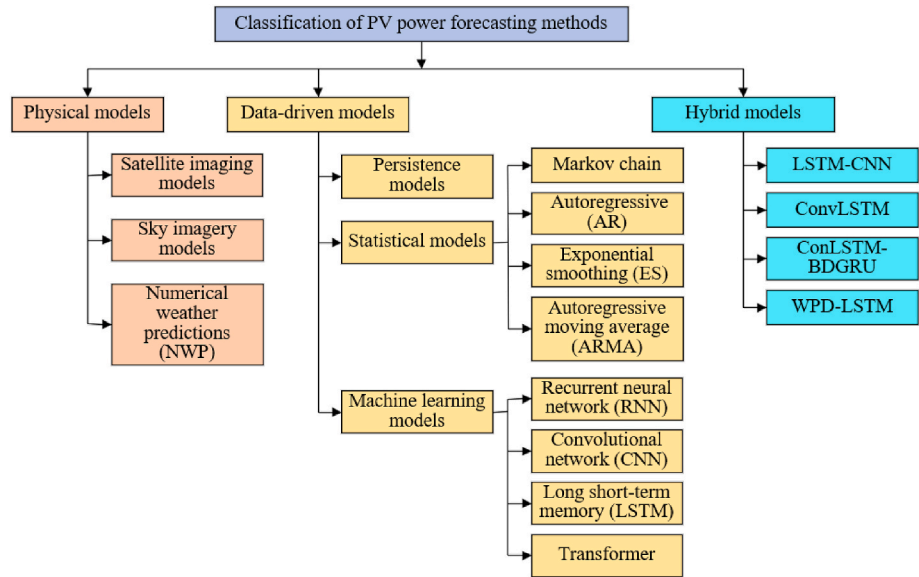


Fig. 2. Classification of PV power forecasting approaches.

others. These approaches can achieve reliable performance as they continuously optimize parameters with historical data during the training phase [16]. Nonetheless, these statistical models still have shortcomings in learn nonlinear relationships [17].

Over the past years, machine learning methods have achieved superiority in learning complex nonlinear features and are utilized in PV power forecasting [18,19]. Support Vector Machines (SVM) [20] is simple to use in forecasting wind and solar energy resources and allow many situations and predictions have a certain degree of interpretability. Recurrent Neural Network (RNN) [21,22] has significant advantages in the field of sequence modeling, with variants like Long Short-term Memory (LSTM) [23] and Gated Recurrent Units (GRU) [24], mitigating the issues of vanishing or exploding gradients in traditional RNNs. Convolutional Neural Network (CNN) [25,26] is good at capturing spatial information among various variables. However, these traditional machine learning models encounter challenges in capturing long-term intervariable dependencies.

Hybrid models integrate physical, statistical, and machine learning

methods to learn temporal and spatial patterns of historical data [27]. Some studies combine CNN and RNN variants to explore spatiotemporal information of historical PV power data [28–30]. The commonly used hybrid models combining CNN and RNN for photovoltaic power prediction include: LSTM-CNN [31], GRU-CNN [32], CNN-LSTM [33], Wavelet Packet Decomposition (WPD) with LSTM [34], and ConLSTM-BDGRU [35]. In the field of PV power forecasting, these hybrid models achieve promising predictions.

Nomenclature			
PV	Photovoltaic	RNN	Recurrent Neural Network
CNN	Convolutional Neural Network	NWP	Numerical Weather Prediction
RWKV	Receptance Weighted Key Value	FFT	Fast Fourier Transform
TCN	Temporal Convolutional Network	MSRWKV-2DTCN	Multi-scale RWKV with 2-Dimensional TCN
AR	Autoregressive	ARMA	Autoregressive Moving Average

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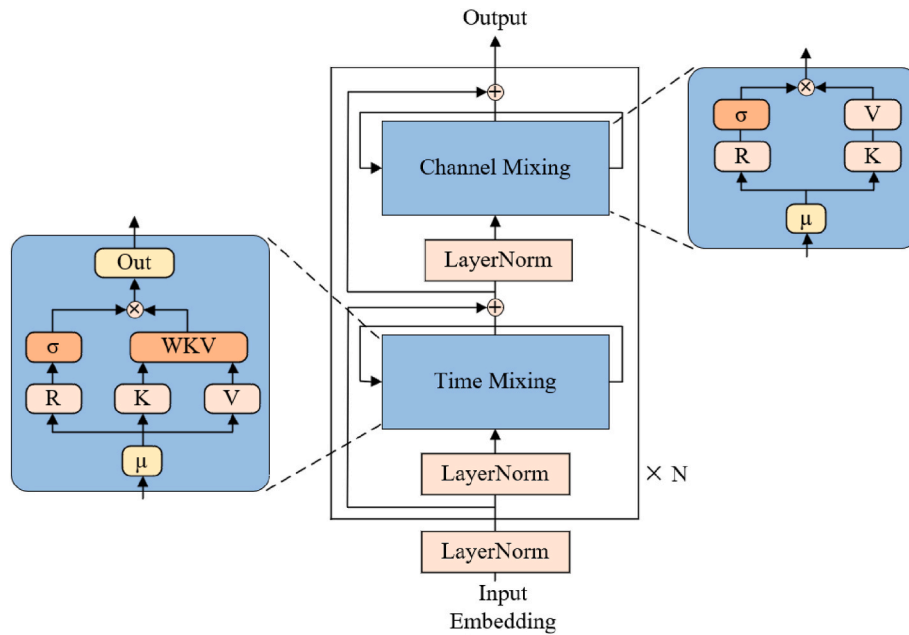


Fig. 3. The architecture of RWKV.

(continued)

ES	Exponential Smoothing	GRU	Gated Recurrent Units
SVM	Support Vector Machines	DKASC	Desert Knowledge Australia Solar Centre
WPD	Wavelet Packet Decomposition	LSTM	Long Short-term Memory
AP	Active Power	CPA	Current Phase Average
WS	Wind Speed	MWS	Max Wind Speed
hPa	Hectopascal	APre	Air Pressure
GHR	Global Horizontal Radiation	PM	Pyranometer
WTC	Weather Temperature Celsius	WDR	Weather Daily Rainfall
TP1	Temperature Probe 1	TP2	Temperature Probe 2
m/s	Meter per second	NRMSE	Normalized Root Mean Square Error
°C	Celsius	w/m ²	Watts per square meter
mm	Millimeter	°	Degree
WD	Wind Direction	A	Ampere
MSE	Mean Squared Error	MAE	Mean Absolute Error
DSN	Dual Stream Network	KW	Kilowatt
MBE	Mean Bias Error	RMSE	Root Mean Square Error
ELM	Extreme Learning Machine	GCN	Graph Convolution Network

Traditional machine learning methods, such as RNN and its variants, suffer from long-term dependency issues, non-parallelizable training, and insensitivity to sequence length, leading to information loss or redundancy. To tackle these challenges, Vaswani et al. [36] proposed the Transformer, which replaces recurrent neural units with attention mechanisms. The Transformer has been adapted to solve PV power forecasting problems in some studies. Sherzobek et al. [37] used the encoder architecture of the Transformer for PV power generation forecasting and achieved low error. Phan et al. [38] introduced the combination of Transformer architecture and XGBoost to enhance PV power forecasting. Wang et al. [1] combined the Transformer architecture with a causal convolution network for PV power forecasting. However, the Transformer suffers from high computational complexity, especially for long sequences [39,40]. Recently, Peng et al. [41] proposed the RWKV, which leverages a linear attention mechanism and achieves efficient inference of RNN with the parallelizable training of the Transformer.

Current studies of PV power forecasting have limitations in simultaneously capturing both the complex periodicity and nonlinear

interdependencies of historical data, which hinders accurate PV power forecasting. Moreover, although RWKV has emerged as an effective solution in processing lengthy sequence [42–45], it has limitations in capturing long-term dependencies as it only pays attention to one previous step and the current state, while the standard Transformer takes full information by the dot-product attention. Additionally, the channel-mixing block in RWKV falls short in extracting complex interdependencies among various variants. Therefore, we introduce a novel hybrid model called MSRWKV-2DTCN designed for short-term PV power forecasting, which combines FFT and a 2D TCN with the RWKV architecture. The FFT can identify multiple periodic patterns in time series data, helping the time mixing block of the RWKV model to focus on critical information. 2D TCN with multi-periodicity discovered by FFT can explore multi-scale interdependencies among subseries. The main innovations are highlighted as follows:

- (1) Current PV power forecasting studies face challenges in effectively capturing the complex periodicity of historical data. To this end, we use FFT to transform historical data from the time domain to the frequency domain for analysis, allowing the identification of various periodic patterns within the historical data.
- (2) A novel multi-scale time mixing block is introduced, integrating the canonical time mixing block of RWKV with the periods identified by FFT. This proposed multi-scale time mixing block extracts dependencies across multiple time scales from historical data and adaptively aggregates these dependencies.
- (3) Given the significant influence of meteorological conditions on PV power and complex interdependencies within historical data, we enhance the RWKV model by replacing its channel-mixing block with a multi-scale 2D TCN. This modification enables the model to effectively learn intricate periodic patterns and dependencies introduced by varying weather conditions.

The remainder of this paper is structured as: Section 2 discusses the foundational concepts of the RWKV architecture; Section 3 illustrates datasets utilized in experiments and the framework of the proposed MSRWKV-2DTCN; Section 4 presents a comprehensive account of the experiments conducted; Section 5 summarizes this article.

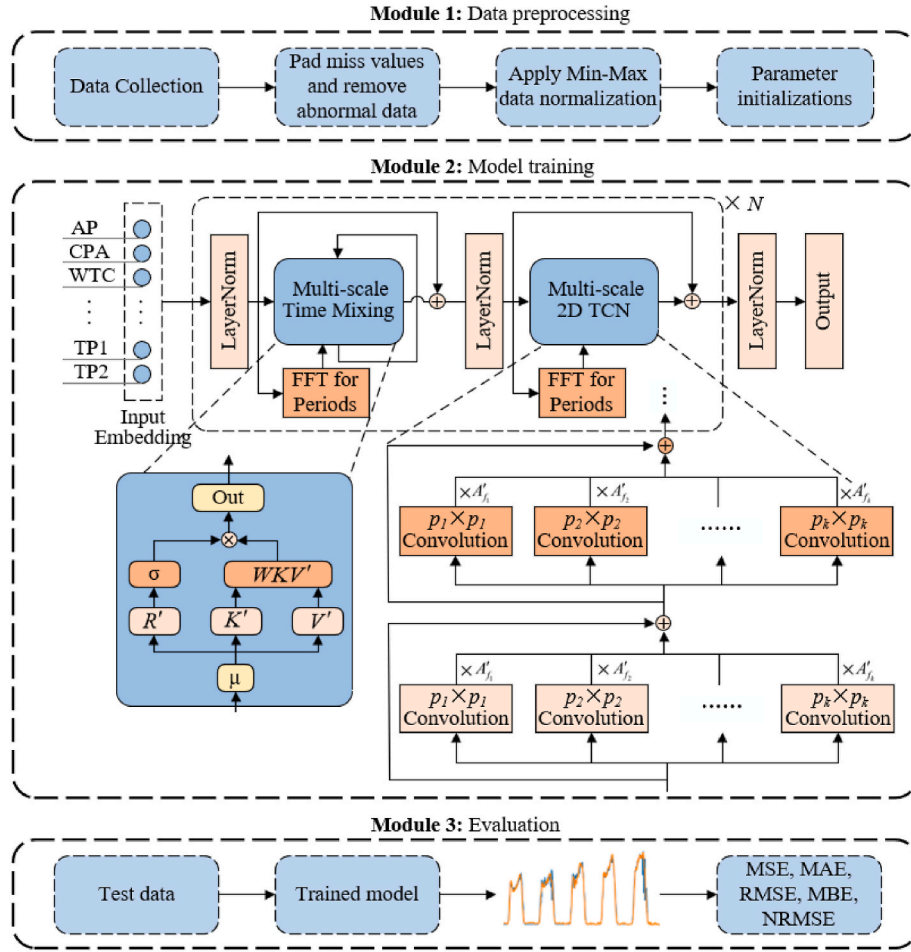


Fig. 4. The architecture of the MSRWKV-2DTCN for PV power forecasting.

Table 1

Configuration of the two PV systems.

Technical specification	DKASC-CA	DKASC-DG
Generation capacity	105.9 kW	1058.4 kW
Array structure	Fixed: Roof Mount	Fixed: Ground Mount
Number of panels	326	3360
PV technology	mono-Si	poly-Si
Array tilt/Azimuth	Tilt = 10, Azi = 40 (E of N)	Tilt = 15, Azi = 0 (Solar North)
Installation completed	3 Mar 2016	3 Mar 2016

2. A prior knowledge of the RWKV architecture

The RWKV [41] architecture, as illustrated in Fig. 3, features a composition of stacked residual blocks, each integrating a time-mixing block alongside a channel-mixing block. Utilizing a linear attention mechanism, RWKV facilitates parallel computations during training while preserving consistent computational and memory efficiency during inference.

Time-mixing block The token shift strategy and hidden states are designed to augment the modelling of interdependencies and patterns within a sequence, replacing quadratic calculation of attention mechanism in Transformer. The main formulations for this block are shown as follows:

$$r_t = W_r \cdot (\mu_r x_t + (1 - \mu_r) x_{t-1}) \quad (1)$$

$$k_t = W_k \cdot (\mu_k x_t + (1 - \mu_k) x_{t-1}) \quad (2)$$

$$v_t = W_v \cdot (\mu_v x_t + (1 - \mu_v) x_{t-1}) \quad (3)$$

$$wkv_t = \frac{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} v_i + e^{u+k_t} v_t}{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} + e^{u+k_t}} \quad (4)$$

$$o_t = W_o \cdot (\sigma(r_t) \odot wkv_t) \quad (5)$$

where the computation of wkv_t with constant computational and memory complexity replaces the $Attention(Q, K, V)$ in Transformers. The parameters u_r , u_k and u_v denote the interpolation weight of the token shift strategy. w and u are scalars. σ represents the sigmoid function.

Channel-mixing block is formulated as:

$$r_t = W_r \cdot (\mu_r x_t + (1 - \mu_r) x_{t-1}) \quad (6)$$

$$k_t = W_k \cdot (\mu_k x_t + (1 - \mu_k) x_{t-1}) \quad (7)$$

$$o_t = \sigma(r_t) \odot (W_v \cdot ReLU(k_t)^2) \quad (8)$$

where the squared $ReLU$ activation function is adopted.

3. Methodology

The framework of MSRWKV-2DTCN for PV power forecasting is illustrated in Fig. 4. To identify various periodic patterns, the historical time series data is first transformed to the frequency domain using FFT. Subsequently, the identified periods and historical data are inputted to

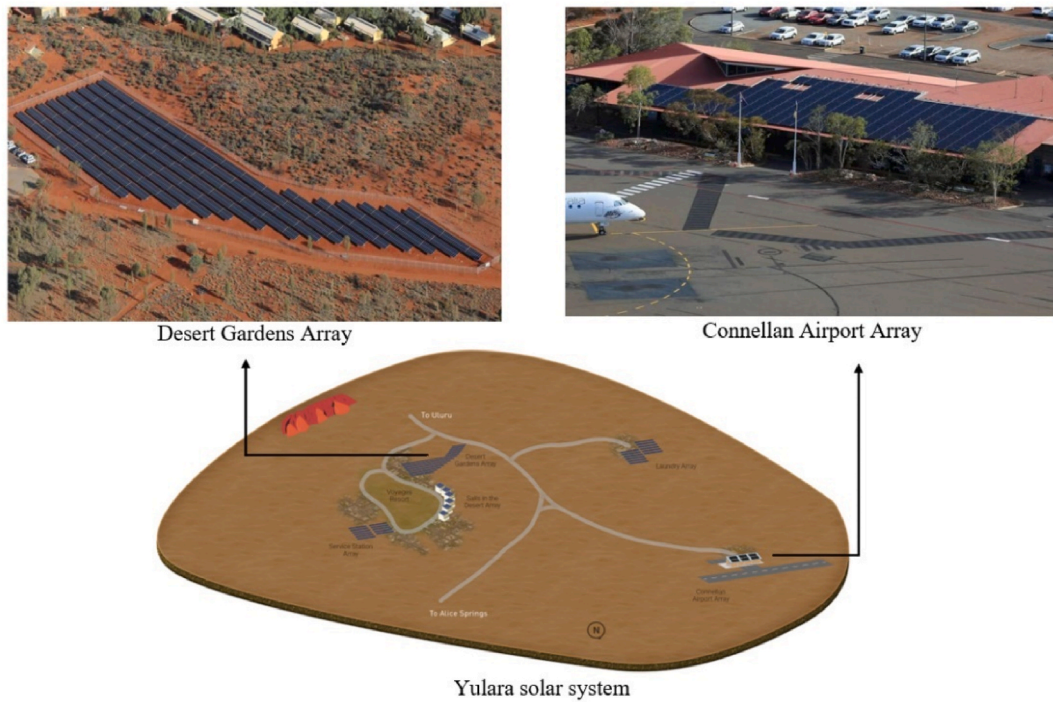


Fig. 5. Two selected PV systems in Yulara solar system.

Table 2

Each attribute of two PV systems and meteorological information.

	Attribute	Min	25 %	50 %	75 %	Max	Mean	STD	Unit
DKASC-DG	AP	−0.25	−0.17	0.31	226.5	791.55	121.08	170.39	KW
	CPA	0.00	4.43	7.30	326.69	1127.85	177.37	242.53	A
DKASC-CA	AP	−0.04	−0.04	0.01	45.45	102.13	22.00	30.54	KW
	CPA	0.58	0.62	0.83	66.79	151.74	32.73	44.52	A
Meteorological information	WS	0.00	1.10	1.85	2.77	12.88	2.03	1.21	m/s
	WTC	0.10	16.10	22.30	28.17	44.82	22.15	8.27	°C
	GHR	−15.00	−3.37	2.25	510.69	1406.09	252.54	349.65	w/m ²
	WD	0.00	130.37	152.47	197.03	349.53	167.35	60.29	°
	WDR	0.00	0.00	0.00	0.00	0.12	0.00	0.00	mm
	MWS	0.00	2.10	3.50	5.40	24.30	3.86	2.27	m/s
	APre	940.45	952.43	956.30	960.51	972.71	956.64	5.49	hPa
	PM	−14.13	−3.17	2.26	549.70	1402.57	268.54	368.52	w/m ²
	TP1	30.39	35.54	39.96	49.14	82.16	43.33	10.34	°C
	TP2	30.37	35.40	39.84	50.02	80.61	43.46	10.46	°C

the multi-scale time mixing block and the multi-scale 2D TCN block to learn intricate periodic patterns and relationships. Each module is detailed in the following sections.

3.1. Datasets and the segmented filling strategy

Experiments are conducted on real-world datasets collected from Yulara solar system at the Desert Knowledge Australia Solar Centre (DKASC) [46], in Australia. To thoroughly evaluate the adaptability and performance of our model, we select two different PV systems, including Desert Gardens Array (DKASC-DG) and Connellan Airport Array (DKASC-CA). Table 1 and Fig. 5 show the configuration details of the two selected PV systems. The two datasets include historical PV active power and meteorological information recorded by the Desert Gardens Weather Station. Each dataset consists of timestamp, Active Power (AP), Current Phase Average (CPA), and meteorological information such as Weather Daily Rainfall (WDR), Wind Speed (WS), Wind Direction (WD), Max Wind Speed (MWS), Air Pressure (APre), Global Horizontal Radiation (GHR), Pyranometer (PM), Weather Temperature Celsius (WTC), Temperature Probe 1 (TP1), and Temperature Probe 2 (TP2). The time

span of the two datasets is from 2021/1/1 0:00:00 to 2021/12/31 23:55:00, with a resolution of 5 min. Statistics about the PV systems and meteorological data is listed in Table 2. Fig. 6 shows historical data of DKASC-DG dataset and meteorological information for January.

The “Active Power” represents the output power of PV systems. As shown in Table 2, the minimum Active Power value is negative. This is due to the PV system has basic power consumption when the solar power generation equipment does not generate electricity at night. There is no need to predict the basic power consumption of the device, so we convert negative values to zero following the way of Khan et al. [27].

Furthermore, the two datasets contain missing values in two scenarios: occasional missing values scattered throughout and consecutive missing values. Missing data can result in partial information loss, potentially affecting the model’s accuracy and generalization capability. If the pattern of missing data is related to other data, the model may make erroneous inferences due to this bias. This effect can be more pronounced, especially in cases with many consecutive missing data points. To address missing values in the two datasets, we designed a segmented filling strategy: for cases where the consecutive missing values are fewer than 12 points (within 1 h), missing time points are

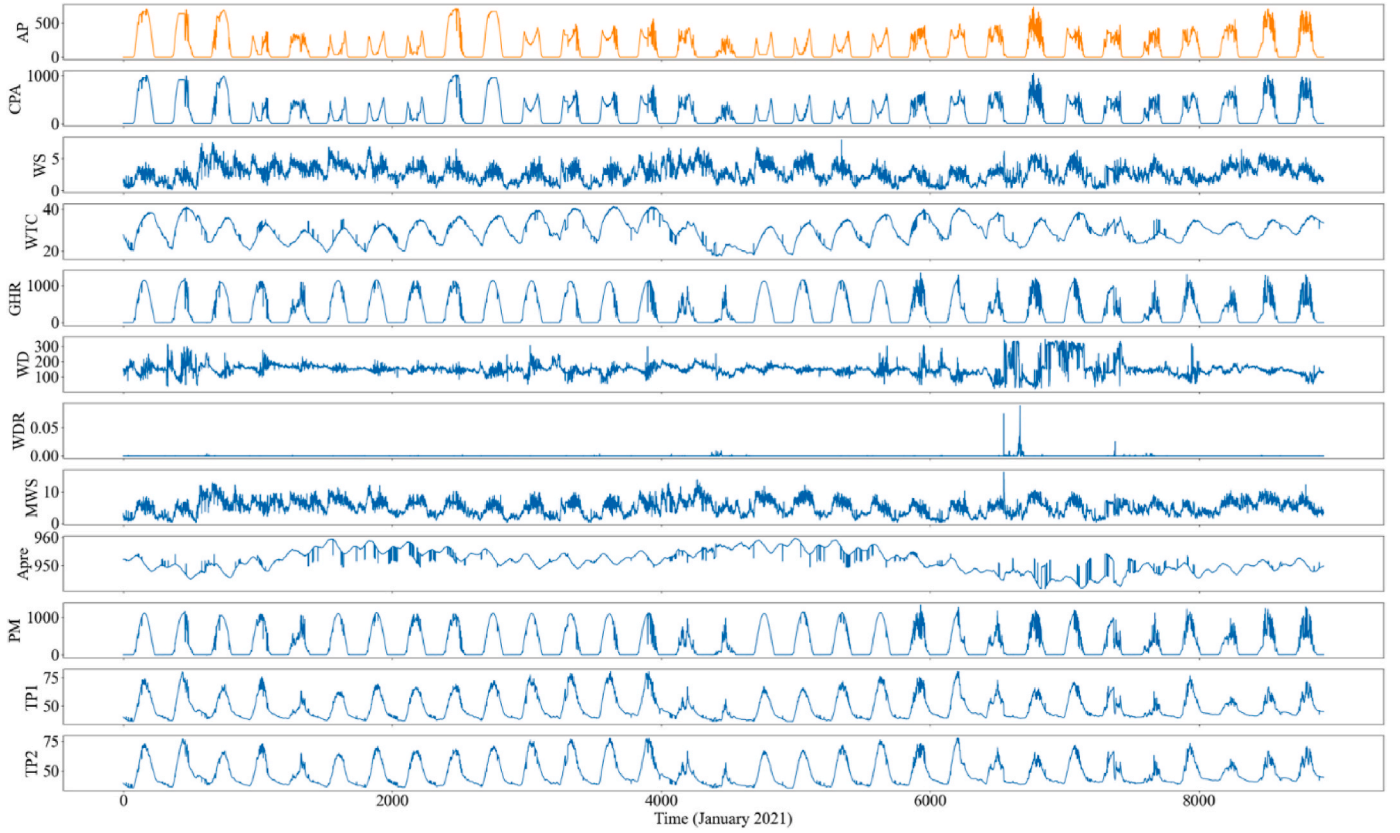


Fig. 6. Historical data of DKASC-DG dataset and meteorological information for January.

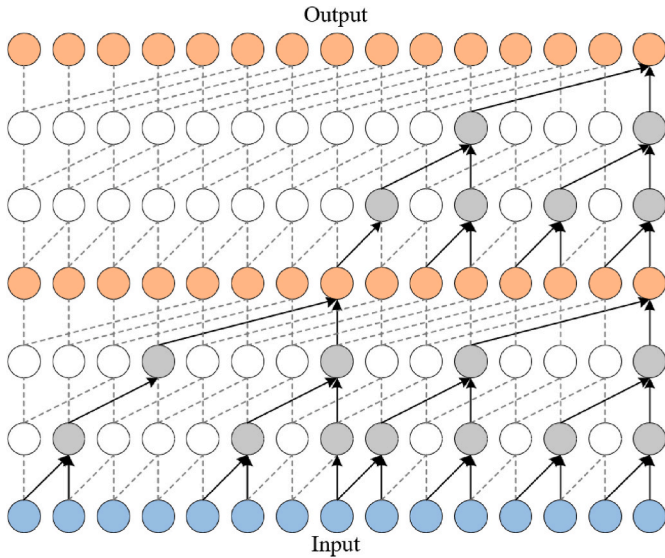


Fig. 7. Two stacked dilated causal convolution layers with kernel size 2 and dilation rate [1, 2, 4].

filled by the linear interpolation method; for cases where the consecutive missing values exceed 12 points, the mean of the corresponding time points from the seven days before and after the missing data is used to pad the missing value. The linear interpolation method is formulated as follows:

$$X' = \frac{t_{miss} - t_{pre}}{t_{next} - t_{pre}} \times X_{pre} + \frac{t_{next} - t_{miss}}{t_{next} - t_{pre}} \times X_{next} \quad (9)$$

where X' is the calculated values for the missing point; X_{pre} and X_{next} represent the first available point before and after the missing point, respectively; t_{miss} , t_{pre} , and t_{next} are the time of missing value, X_{pre} , and X_{next} , respectively.

3.2. FFT

Time series in the real world, especially for PV power generation, often exhibit multiple periodic patterns. The selected datasets contain historical PV output power and meteorological information, both of which show significant periodic fluctuations. Meteorological information can display daily, monthly, and yearly variations, which in turn influence the PV output power, adding to its complexity and instability. Understanding these periodic patterns is crucial for building accurate forecasting models, as these patterns significantly affect the overall data trends. Therefore, recognizing and analyzing these periodicities, especially considering the influence of meteorological conditions on the output of PV systems, is essential for improving forecasting accuracy. To achieve this, we employ the FFT technique to transform the time series data to the frequency domain, enabling the identification of various periodic patterns in both historical PV output power and meteorological information as follows:

$$A = \text{Avg}(\text{Amp}(\text{FFT}(X))) \quad (10)$$

$$\{f_1, f_2, \dots, f_k\} = \text{Topk}(A) \quad (11)$$

$$f_s \in \left\{ 1, \dots, \left\lfloor \frac{L}{2} \right\rfloor \right\}$$

where $\text{FFT}(X)$ denotes the process of FFT on multivariate time series $X \in R^{L \times N}$, $\text{Amp}()$ is used to calculate values of amplitudes, $\text{Avg}()$ represents the calculation of the average of N dimensions, and L is the sequence length. As the frequency domain is conjugate, only the

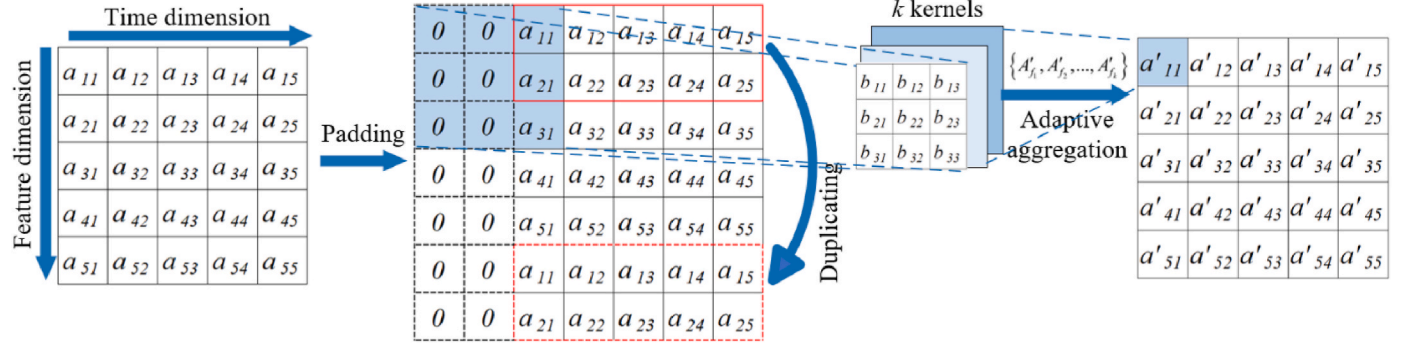


Fig. 8. The procedure of 2D TCN with k different convolution kernels.

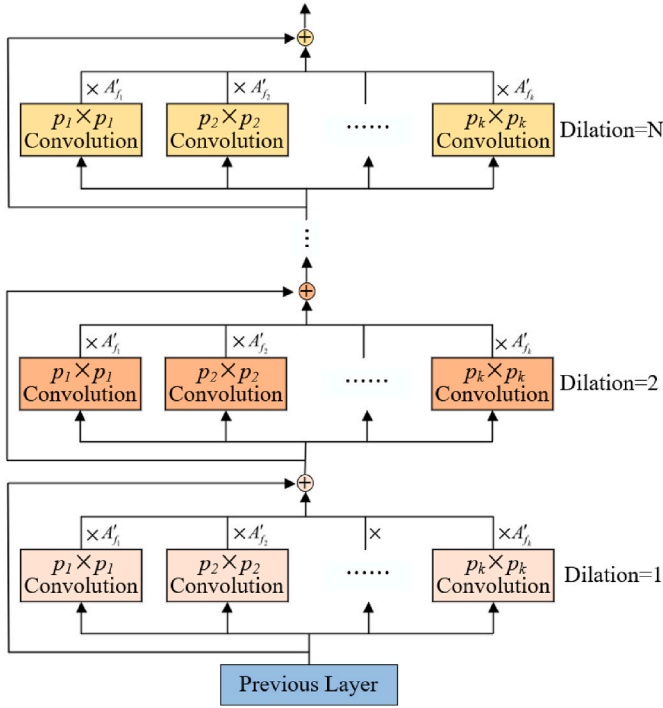


Fig. 9. The architecture of the multi-scale 2D TCN.

Table 3

Range of hyperparameters.

Parameter	Range
Stacked layers	[1, 2, 3, 4]
Value of k	[3, 5, 7, 9]
Hidden size	[128, 256, 512, 1024]

frequencies within the range of $\{1, \dots, [L/2]\}$ is considered. To mitigate frequency domain sparsity and minimize noise from irrelevant high frequencies, we choose the top-k values of $\mathbf{A}$, where k is a hyperparameter. This selection can figure out the top-k significant frequencies $\{f_1, f_2, \dots, f_k\}$ and obtain k amplitudes $\{A_{f_1}, A_{f_2}, \dots, A_{f_k}\}$. The selected frequencies correspond to k important periods $P = \{p_1, p_2, \dots, p_k\}$. These periods are calculated as follows:

$$p_i = \frac{L}{f_i}, i \in \{1, \dots, k\} \quad (12)$$

Table 4

Experimental results of MSRWKV-2DTCN and comparison models.

Dataset	Method	MSE	MAE	MBE	RMSE	NRMSE
DKASC-DG	MSRWKV-2DTCN	0.0043	0.0277	-0.0021	0.0656	0.2806
	DSN	0.0057	0.0411	-0.0054	0.0755	0.3230
	GRU-CNN	0.0142	0.0928	-0.0227	0.1192	0.5098
	CNN-LSTM	0.0167	0.0972	-0.0244	0.1292	0.5529
	Crossformer	0.0074	0.0534	-0.0190	0.0860	0.3681
	Informer	0.0114	0.0685	-0.0245	0.1068	0.4568
	LSTM	0.0325	0.1157	-0.0345	0.1803	0.7713
	TCN	0.0291	0.1012	-0.0311	0.1706	0.7299
DKASC-CA	MSRWKV-2DTCN	0.0047	0.0252	-0.0022	0.0686	0.2720
	DSN	0.0061	0.0393	-0.0051	0.0781	0.3099
	GRU-CNN	0.0155	0.0873	-0.0144	0.1245	0.4940
	CNN-LSTM	0.0153	0.0920	-0.0152	0.1237	0.4908
	Crossformer	0.0079	0.0501	-0.0045	0.0889	0.3526
	Informer	0.0117	0.0637	-0.0092	0.1082	0.4292
	LSTM	0.0303	0.1032	-0.0324	0.1741	0.6906
	TCN	0.0270	0.0955	-0.0302	0.1643	0.6519

3.3. Multi-scale time mixing block

The canonical time mixing block of RWKV leverages a linear attention mechanism, facilitating the parallelizable training of the Transformer. However, it encounters challenges in capturing complex dependencies because it only considers one previous step and the current state. The output of solar power systems is affected by meteorological factors and typically exhibits complex cyclic patterns. Therefore, incorporating essential periodic information into the forecasting model is advantageous. We designed a multi-scale time mixing block that expands the receptive field, preserves important periodic information, and learns multi-scale dependencies. The calculation of the proposed multi-scale time mixing block is expressed as follows:

$$\{A'_{f_1}, A'_{f_2}, \dots, A'_{f_k}\} = \text{Softmax}(A_{f_1}, A_{f_2}, \dots, A_{f_k}) \quad (13)$$

$$r'_t = W_r \cdot \left(\mu_r x_t + (1 - \mu_r) \left(x_{t-1} + \sum_{i=1}^k A'_{f_i} x_{t-p_i} \right) \right) \quad (14)$$

$$k'_t = W_k \cdot \left(\mu_k x_t + (1 - \mu_k) \left(x_{t-1} + \sum_{i=1}^k A'_{f_i} x_{t-p_i} \right) \right) \quad (15)$$

$$v'_t = W_v \cdot \left(\mu_v x_t + (1 - \mu_v) \left(x_{t-1} + \sum_{i=1}^k A'_{f_i} x_{t-p_i} \right) \right) \quad (16)$$

$$wkv'_t = \frac{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k'_i} v'_i + e^{u+k'_i} v'_i}{\sum_{i=1}^{t-1} e^{-(t-1-i)w+K_i} + e^{u+K_i}} \quad (17)$$

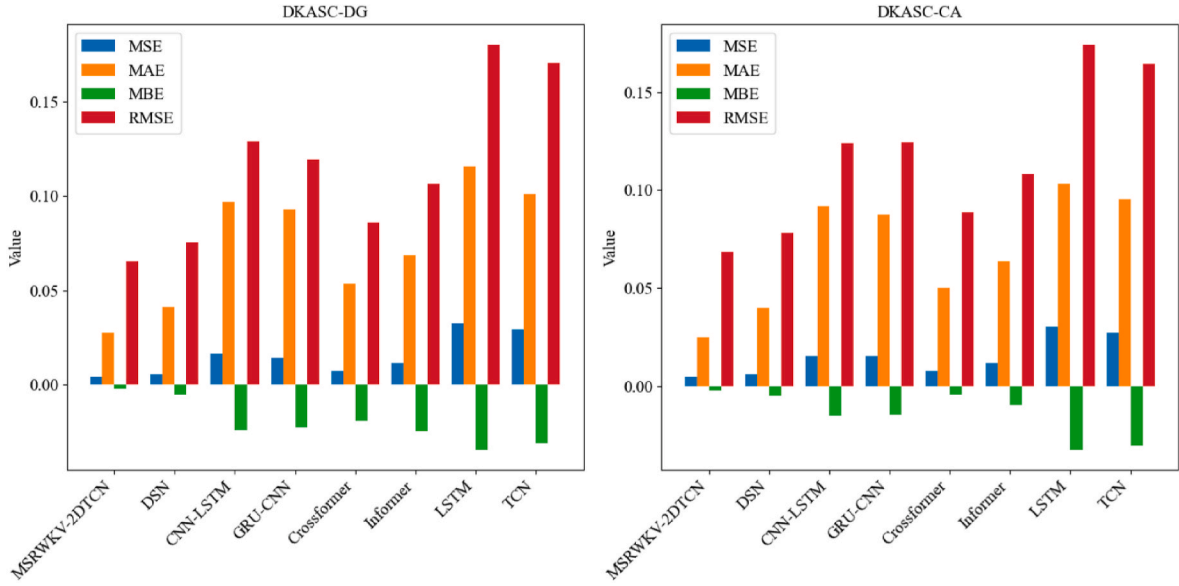


Fig. 10. Performance comparison of MSRWKV-2DTCN and other models.

$$o_t = W_o \cdot (\sigma(r'_t) \odot wkv'_t) \quad (18)$$

where $\{A_{f_1}, A_{f_2}, \dots, A_{f_k}\}$ are k amplitudes corresponding to the top- k significant frequencies $\{f_1, f_2, \dots, f_k\}$, the parameters u_r , u_k and u_v denote the interpolation weight of the token shift strategy, w and u are scalars, and the symbol σ represents the sigmoid function.

Inspired by previous studies like Timesnet [40] and Autoformer [47], the relative importance of selected frequencies and their corresponding periods can be represented by the amplitudes $\{A_{f_1}, A_{f_2}, \dots, A_{f_k}\}$. In Equation 13, the top- k selected amplitudes are processed by the *Softmax* function, serving as weights for each period scale. This enables the adaptive aggregation of multiple periods during the calculation of r'_t , k'_t , and v'_t . Consequently, the proposed multi-scale time mixing block not only considers the current and previous states but also integrates significant periodic information across different scales.

3.4. Multi-scale 2D TCN

The output of PV systems is significantly affected by meteorological factors, making it crucial to understand the complex interdependencies within the data for accurate PV power forecasting. Although the RWKV model is effective in sequential modeling, it struggles to capture intricate relationships between different time series features due to the simplicity of its channel-mixing block. To address this limitation, we replace it with a multi-scale 2D TCN block that incorporates the periods identified by FFT as convolution kernels. TCN [48] is an effective architecture designed for sequence modeling. Unlike RNNs that process sequences sequentially, the convolutional operations of the TCN can be computed in parallel, enhancing efficiency when processing long sequences. Additionally, TCN mitigates the vanishing or exploding gradients commonly encountered in RNNs by expanding the receptive field of convolutions, facilitating the capture of long-range dependencies. Furthermore, TCN leverages dilated causal convolution, as illustrated in Fig. 7, which can increase the receptive field by introducing gaps between the elements of the kernel.

TCN typically use one-dimensional (1D) convolution to process time series data. The 1D convolution operation applies the convolution kernel to each time step, capturing both local patterns and long-term dependencies. However, 1D convolution operations cannot adequately capture the complex dependencies between PV power and meteorological information, as they apply the convolution kernel independently to

each time series and do not directly capture relationships between different dimensions in a multivariate time series. Therefore, we introduce a multi-scale 2D TCN with 2D convolution to better capture the complex dependencies between PV power and external factors. To avoid information leakage, we apply zero padding to the left of the time dimension with the padding length $(K-1) \times D$, where K represents size of kernel, and D denotes the dilation rate. Additionally, the first $(K-1) \times D$ features are duplicated and placed after the last feature as padding data for the feature dimension. Fig. 8 illustrates the procedure of 2D TCN.

In the proposed multi-scale 2D TCN, the lengths of selected k periods $\{p_1, p_2, \dots, p_k\}$ are used as kernel sizes for 2D convolution. These k different representations, generated by k kernels, are adaptively aggregated using the weights $\{A'_{f_1}, A'_{f_2}, \dots, A'_{f_k}\}$:

$$X^{M \times N} = X^{M \times N} + \sum_{i=1}^k A'_{f_i} \times X_i^{M \times N} \quad (19)$$

where $X^{M \times N}$ denotes the output of previous layer, $X_i^{M \times N}$ represents the result of the convolutional operation with kernel size $p_i \times p_i$, and A'_{f_i} denotes the relative weight of the frequency f_i . The architecture of the multi-scale 2D TCN is illustrated in Fig. 9.

Notably, the proposed multi-scale 2D TCN effectively captures both intraperiod and interperiod variations. By incorporating multiple periods as convolutional kernels, the model can identify the intraperiod patterns or relationships within different period. Additionally, the use of dilated convolutions allows kernels to cover a larger receptive field by skipping time steps, which enables the model to capture interperiod relationships. This combination of periodicity-based kernels and dilated convolutions enhances the model's ability in learning temporal relationships in time series data, ensuring that both intraperiod and interperiod interdependencies are accurately captured.

4. Experimental results

4.1. Data normalization

In this study, the data is scaled to the range [0,1] with the Min-Max normalization. The distribution of PV output and changes in meteorological factors is not strictly normal but closer to cyclical variation. Moreover, the range of values for different features varies widely.

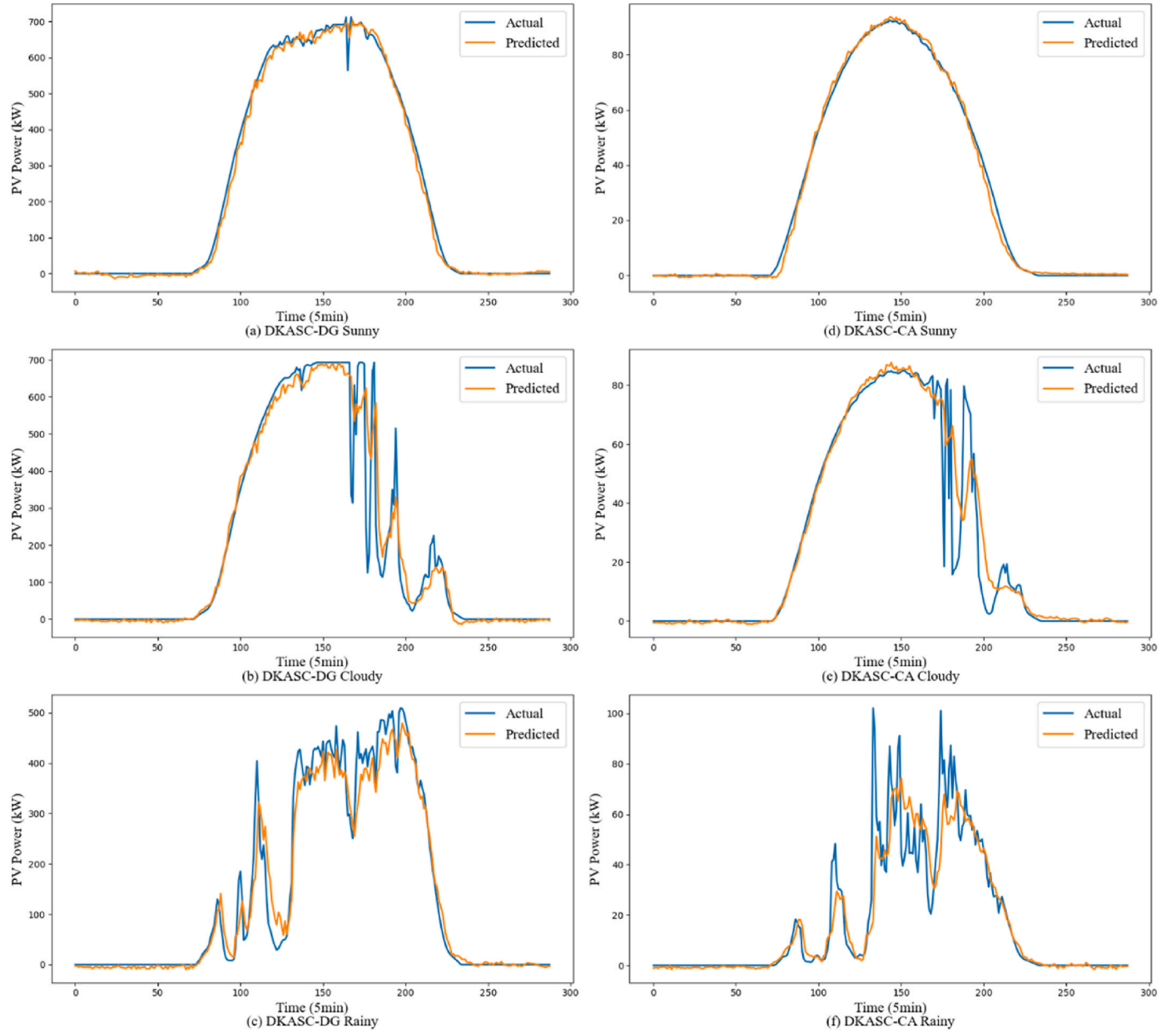


Fig. 11. Forecasting results of MSRWKV-2DTCN and actual values under different weather conditions.

Table 5
Ablation study of MSRWKV-2DTCN.

Datasets	Metric	DKASC-DG	DKASC-CA
MSRWKV-2DTCN	MSE	0.0043	0.0047
	MAE	0.0277	0.0252
	MBE	-0.0021	-0.0022
w/o MSRWKV	MSE	0.0059	0.0052
	MAE	0.0437	0.0458
	MBE	-0.0040	-0.0037
w/o 2DTCN	MSE	0.0075	0.0071
	MAE	0.0518	0.0503
	MBE	-0.0082	-0.0065
RWKV	MSE	0.0092	0.0085
	MAE	0.0622	0.0641
	MBE	-0.0102	-0.0080

Therefore, Min-Max normalization may be more appropriate when standardizing data because it does not rely on assumptions about the distribution of the data and is able to preserve the relative size relationships of the data, eliminate scale differences between features. The formula of the Min-Max normalization is described as:

$$x_i^n = \frac{x_i^n - x_{\min}^n}{x_{\max}^n - x_{\min}^n} \quad (20)$$

where x_i^n denotes the i -th point of the n -th feature; $x_{\min}^n$ and $x_{\max}^n$ represent the minimum and maximum values of the n -th feature, respectively; x_i^n is the normalized value.

4.2. Evaluation indices

The performance of involved models is evaluated via Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Normalized Root Mean Square Error (NRMSE), and Mean Bias Error (MBE) metrics.

Table 6
Comparatively analysis for each season.

Method	Season	MSE	MAE	MBE	RMSE	NRMSE
MSRWKV-2DTCN	Spring	0.0031	0.0283	-0.0022	0.0557	0.2063
	Summer	0.0034	0.0306	-0.0063	0.0583	0.2270
	Autumn	0.0042	0.0340	-0.0061	0.0648	0.3480
	Winter	0.0032	0.0297	-0.0071	0.0566	0.2630
DSN	Spring	0.0048	0.0497	-0.0086	0.0693	0.2567
	Summer	0.0047	0.0412	-0.0102	0.0686	0.2669
	Autumn	0.0062	0.0496	-0.0091	0.0787	0.4228
	Winter	0.0045	0.0487	-0.0098	0.0671	0.3119
CNN-LSTM	Spring	0.0067	0.0552	-0.0073	0.0819	0.3033
	Summer	0.0058	0.0609	-0.0088	0.0762	0.2965
	Autumn	0.0086	0.0735	-0.0103	0.0927	0.4979
	Winter	0.0062	0.0611	-0.0075	0.0787	0.3661
GRU-CNN	Spring	0.0124	0.0921	-0.0058	0.1114	0.4127
	Summer	0.0137	0.0769	-0.0065	0.1170	0.4556
	Autumn	0.0154	0.0947	-0.0098	0.1241	0.6663
	Winter	0.0118	0.0936	-0.0067	0.1086	0.5051
Crossformer	Spring	0.0151	0.1030	-0.0074	0.1229	0.4554
	Summer	0.0158	0.0982	-0.0055	0.1257	0.4893
	Autumn	0.0173	0.1064	-0.0114	0.1315	0.7062
	Winter	0.0160	0.0969	-0.0085	0.1265	0.5881
Informer	Spring	0.0089	0.0623	-0.0082	0.0943	0.3496
	Summer	0.0092	0.0716	-0.0083	0.0959	0.3734
	Autumn	0.0107	0.0813	-0.0127	0.1034	0.5554
	Winter	0.0085	0.0722	-0.0079	0.0922	0.4287

$$MSE = \frac{1}{N} \sum_{i=1}^N (x'_i - x_i)^2 \quad (21)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |x'_i - x_i| \quad (22)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (x'_i - x_i)^2} \quad (23)$$

$$NRMSE = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (x'_i - x_i)^2}}{X_{mean}} \quad (24)$$

$$MBE = \frac{1}{N} \sum_{i=1}^N (x'_i - x_i) \quad (25)$$

where N denotes the number of samples; x_i and x'_i represents the actual value and the forecasting value, respectively; X_{mean} is the mean of actual values.

MSE, MBE, and MAE evaluate the difference between predicted and actual values. NRMSE is the RMSE normalized to a range, mean, or other statistical property of the actual values. This makes it dimensionless and easier to compare across different datasets. Each metric provides a different perspective on model performance and can be used together to gain a comprehensive understanding of prediction accuracy and error characteristics.

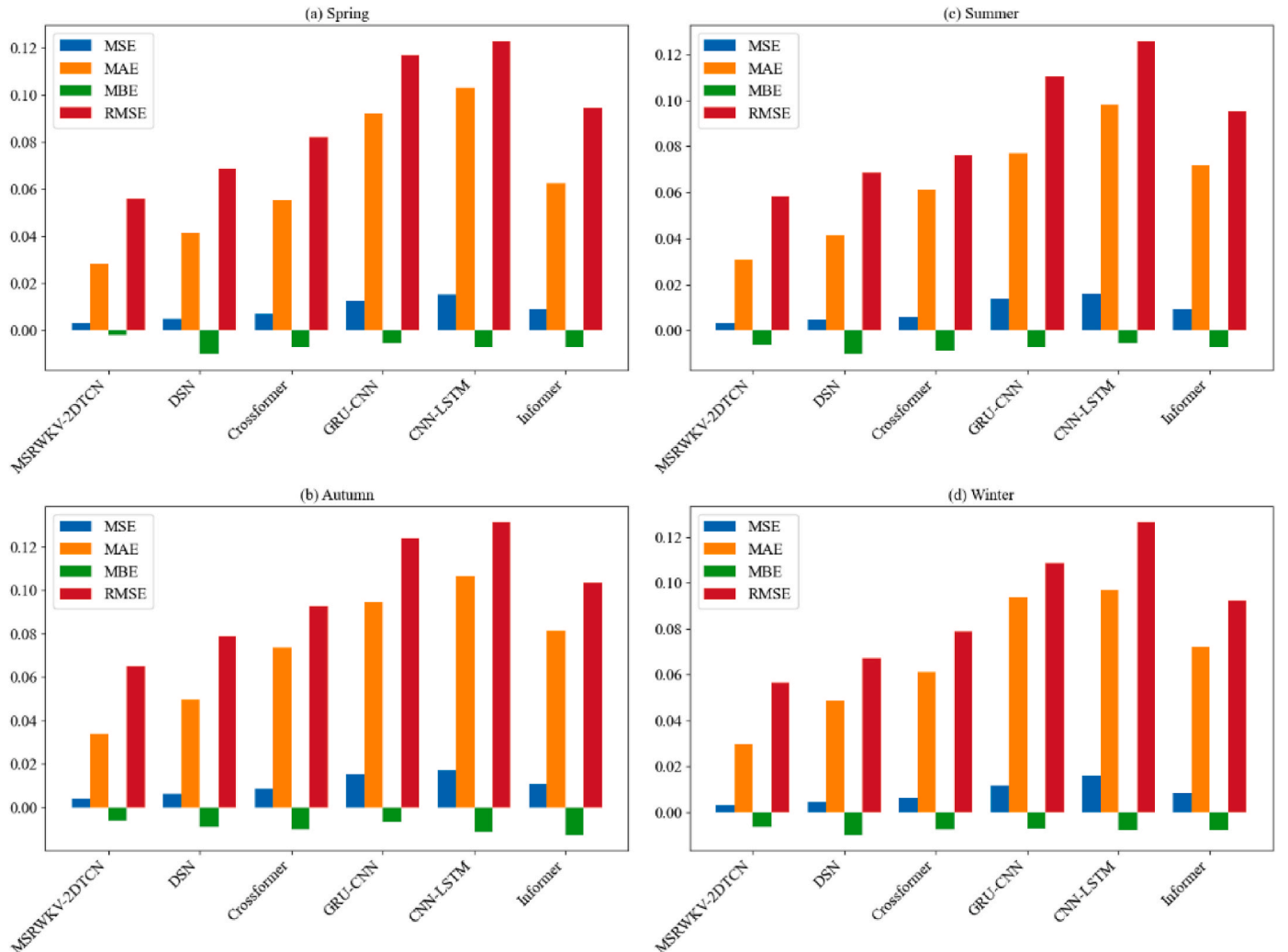


Fig. 12. Performance comparison for each season.

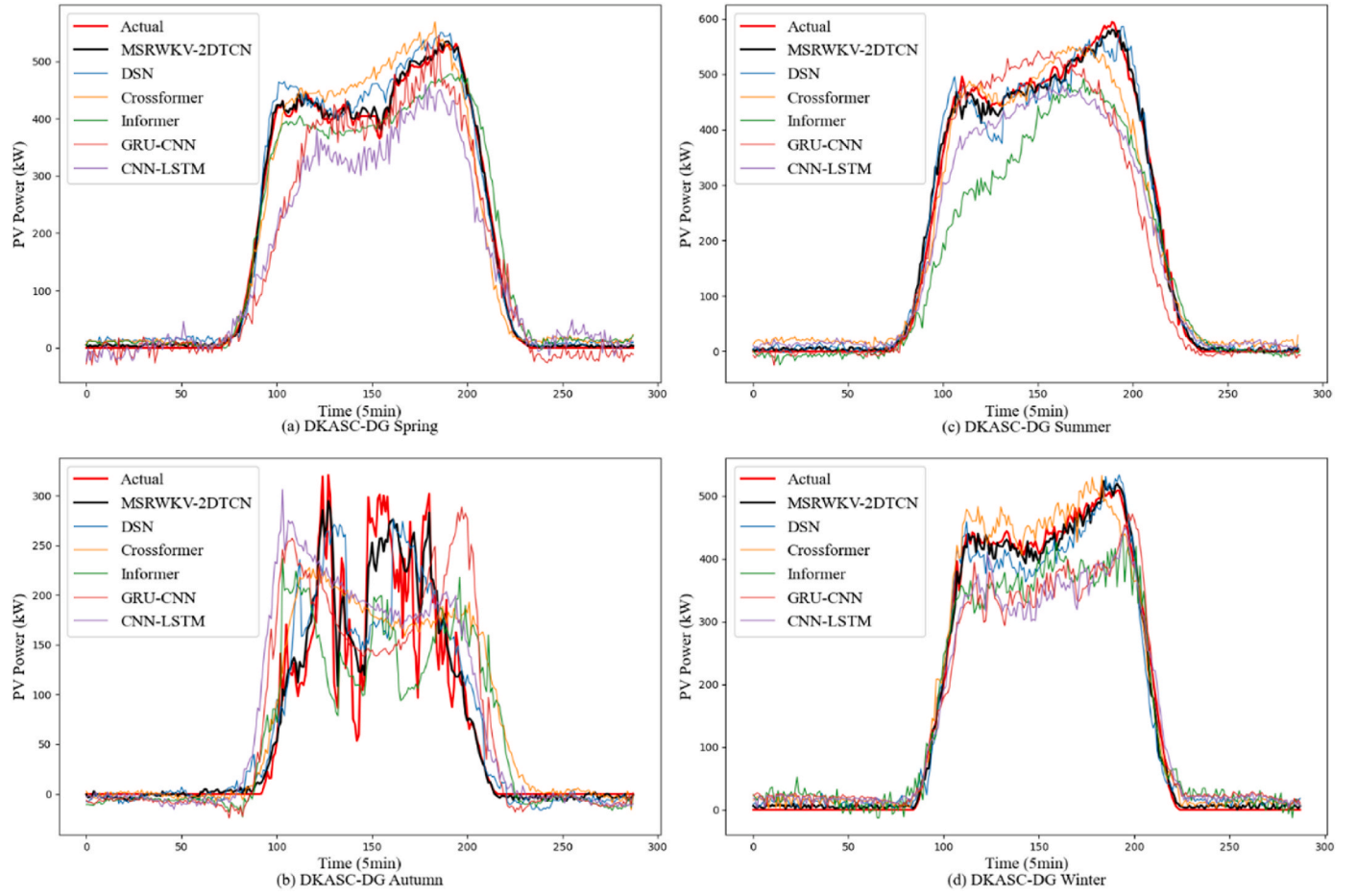


Fig. 13. Prediction results for each season of DKASC-DG dataset.

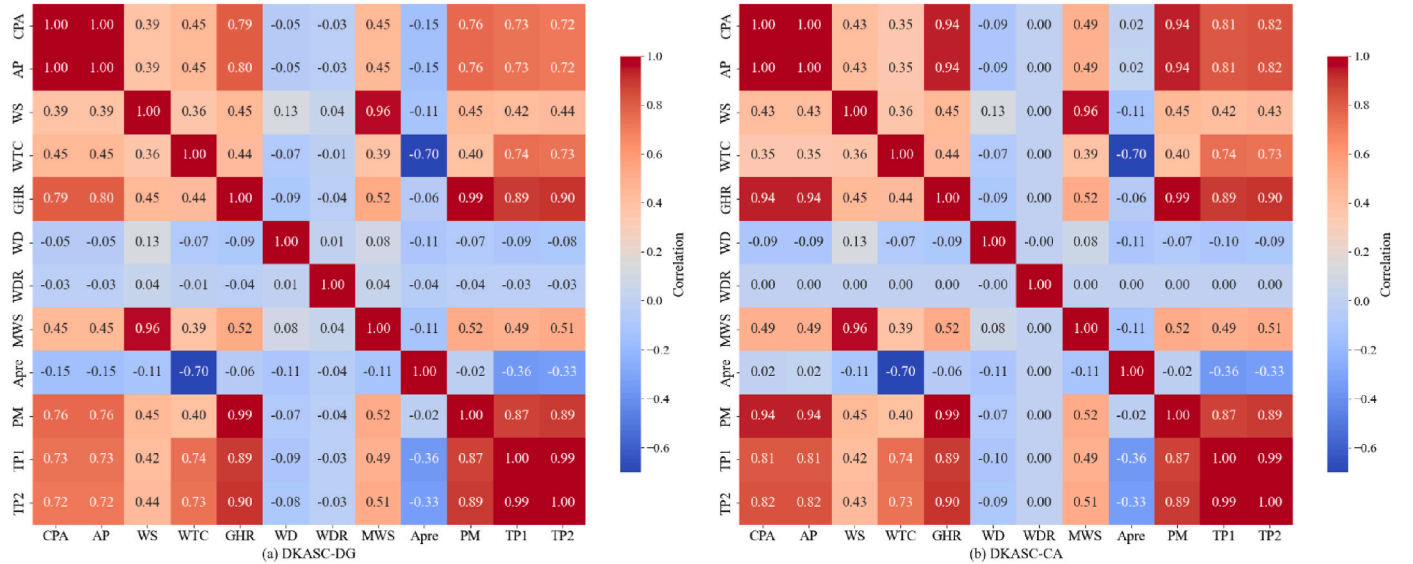


Fig. 14. The correlation matrix heatmaps of two datasets: (a) DKASC-DG and (b) DKASC-CA.

4.3. Experimental setting

The ADAM optimizer and L2 loss function are used in training. The look-back window for MSRWKV-2DTCN and comparison methods is set to 288 (1 day), and the forecasting horizon is 12 (1 h). The two datasets are split into training, validation, and test sets with the ratio of 7:1:2.

Experiments are carried out with PyTorch deep learning framework on a computer with two NVIDIA GeForce RTX 2080 Ti GPUs. The range of hyperparameters of the MSRWKV-2DTCN are shown in Table 3, and they are tuned based on rolling validation error on the validation dataset. According to experimental results, the hyperparameters for the two datasets are set as: the number of stacked layers = 2, the value of k

Table 7

Experimental results of full set of features versus the reduced set of features.

Dataset	Method	MSE	MAE	MBE	RMSE	NRMSE
DKASC-DG	Full features	0.0043	0.0277	−0.0021	0.0656	0.2806
	Reduced features	0.0042	0.0275	−0.0021	0.0648	0.2773
DKASC-CA	Full features	0.0047	0.0252	−0.0022	0.0686	0.2720
	Reduced features	0.0046	0.0252	−0.0021	0.0678	0.2691

Table 8

Experimental results on other DKASC datasets.

Dataset	Method	MSE	MAE	MBE	RMSE	NRMSE
DKASC-ASA-1A	CNN-LSTM [33]	–	–	–	0.3430	0.1260
	CNN [57]	–	–	–	0.3090	0.1750
	GRU-CNN [32]	0.0216	0.0171	–	0.1468	0.0742
	LSTM-CNN [31]	–	–	–	0.6210	0.2210
	ESNCNN [26]	0.0259	−0.0017	–	0.1610	0.0845
	CNNGRU [26]	0.0771	0.0172	–	0.2776	0.1531
	DSN [27]	0.0052	0.0019	–	0.0721	0.0383
	MSRWKV-2DTCN	0.0041	−0.0014	–	0.0640	0.0320
DKASC-ASA-1B	GRU-CNN [32]	0.0298	0.0235	–	0.1727	0.0923
	ESNCNN [26]	0.0463	0.0029	–	0.2153	0.1466
	CNNGRU [26]	0.1605	−0.0300	–	0.4006	0.3247
	LSTM-CNN [26]	0.1249	0.0948	–	0.3535	0.2116
	CNNLSTM [26]	0.1573	0.0113	–	0.3967	0.2408
	DSN [27]	0.0086	−0.0013	–	0.0928	0.0550
	MSRWKV-2DTCN	0.0063	−0.0009	–	0.0794	0.0439

= 5, and the hidden size = 128. In terms of the multi-scale 2D TCN, the dilation rates are set to [1, 2, 4]. Research and practical applications have shown that this set of dilation rates provides a good balance between model complexity and performance [48–50]. Additionally, the selection of these dilation rates was made with careful consideration of the length of input sequence and the size of convolutional kernels, ensuring that the expanded convolutional kernel does not exceed the length of the input data, which is critical for preserving temporal features and maintaining computational efficiency. Therefore, we adopted these dilation rates directly in our experiments.

4.4. Experimental comparison

The effectiveness of the proposed MSRWKV-2DTCN is assessed by comparing it to various deep learning models. These comparison methods can be categorized into three classes: single models (LSTM [23] and TCN [48]), hybrid models (DSN [27], CNN-LSTM [33], and GRU-CNN [32]), and Transformer-based models (Informer [39] and Crossformer [51]). LSTM and TCN are traditional single models for sequence modeling tasks. DSN is a state-of-the-art method in PV power forecasting, which can learn both spatial patterns and temporal representations by combining GRU and CNN. Informer and Crossformer are

efficient methods for time series forecasting. Informer employs the ProbSparse Self-attention mechanism to mitigate the Transformer's computational load. Crossformer models cross-time and cross-dimension dependencies using a two-stage attention mechanism.

The experimental results of MSRWKV-2DTCN and comparison models are shown in Table 4. The results indicate that hybrid models and Transformer-based models achieve significantly lower errors than single models, since single models have limitations in learning both spatial patterns and temporal dependencies. The best results are shown in bold. MSRWKV-2DTCN achieves lower errors of 0.0043, 0.0277, −0.0021, 0.0656, and 0.2806 for MSE, MAE, MBE, RMSE, and NRMSE, respectively, when compared to other prediction models on the DKASC-DG dataset. Compared with these metrics of the DSN, the proposed MSRWKV-2DTCN exhibits reductions of 24.56 %, 32.60 %, 61.11 %, 13.11 %, and 13.13 % for MSE, MAE, MBE, RMSE, and NRMSE, respectively. The MSRWKV-2DTCN also achieves the best performance over comparison methods on the DKASC-CA dataset. Fig. 10 shows the performance comparison of the involved methods. As the NRMSE values are significantly larger than those of the other metrics, for better visual clarity, NRMSE is not displayed in the figure. It can be observed that MSRWKV-2DTCN, DSN, and Crossformer have achieved strong performance, with MSRWKV-2DTCN performing the best.

Additionally, the output of PV power systems is particularly susceptible to abrupt weather fluctuations, such as cloudy, rainy and other weather changes, which have an impact on the stability of the grid. Therefore, the model's performance under rainy and cloudy weather conditions is very important to aid in grid planning and scheduling. Fig. 11 presents the forecasting results of MSRWKV-2DTCN under different weather conditions: sunny, cloudy, and rainy. On sunny days, the model accurately predicts future power output. On rainy and cloudy days, MSRWKV-2DTCN effectively captures the trends of the output power, which can assist in power grid scheduling and planning.

4.5. Ablation study

To further analyze the contribution of every major component of the proposed MSRWKV-2DTCN, we conducted ablation studies on the DKASC-DG and DKASC-CA datasets. Each component of the MSRWKV-2DTCN is replaced or removed in turn to form different sub-frameworks:

- w/o MSRWKV: Replace the multi-scale time mixing block with the canonical time mixing block of RWKV.
- w/o 2DTCN: Replace the multi-scale 2D TCN with the canonical TCN.
- RWKV: The canonical RWKV architecture.

Table 5 presents the comparison of MSRWKV-2DTCN model and the sub-frameworks on DKASC-DG and DKASC-CA datasets. The results show that the complete MSRWKV-2DTCN model outperforms all other variations across all metrics. Notably, replacing the multi-scale 2D TCN block with the canonical TCN results in higher errors than replacing the multi-scale time mixing block with the canonical time mixing block of

Table 9

General comparison of MSRWKV-2DTCN with other state-of-the-art models.

Year	Author	Method	Resolution	Power	MAE	MSE	RMSE
2019	Li et al. [58]	HIMVO-SVM	30 min	–	0.2805	–	–
		MOV-LSTM			0.3705	–	–
2020	Zhou et al. [59]	SDA-GA-ELM	5 min	4.95 KW	0.2367	–	0.2107
		SDA-ELM			0.2717	–	0.2478
2020	Zang et al. [61]	DenseNet	1 h	5.00 KW	0.1520	0.0810	–
		ResNet			0.1800	0.1280	–
2021	Cheng et al. [60]	GCN	5 min	–	0.1770	–	0.3360
2022	Khan et al. [62]	CNNESN	5 min	–	0.0449	0.0095	0.2604
2023	Khan et al. [27]	DSN	5 min	60.4 KW	0.0394 (avg)	0.0051 (avg)	0.0678 (avg)
2024	This paper	MSRWKV-2DTCN	5 min	1164.3 KW	0.0265 (avg)	0.0045 (avg)	0.0671 (avg)

RWKV, highlighting the effectiveness of the multi-scale 2D TCN in capturing relationships between PV power and external factors. The canonical RWKV architecture shows the highest errors among all configurations, reaffirming the contributions of both the multi-scale time mixing block and the multi-scale 2D TCN. In summary, the ablation study confirms that the proposed components significantly enhance the performance of MSRWKV-2DTCN model for PV power forecasting.

4.6. Performance evaluation in different seasons

As seasonal factors have an impact on the output of PV systems, we conducted experiments on each season to assess the performance of MSRWKV-2DTCN. Extensive experiments were conducted for each season of DKASC-DG and DKASC-CA datasets. Comparatively, hybrid models and Transformer-based models have advantages over single models in capturing temporal features and spatial dependencies. Therefore, single models were excluded from further seasonal analyses. Each dataset was divided into four seasons [27]. The average experimental results of DKASC-DG and DKASC-CA across each season are presented in Table 6 and Fig. 12. In each season, the MSRWKV-2DTCN outperformed other hybrid models and Transformer-based models across evaluation metrics. Compared to the DSN, the proposed model shows average reductions of 31.06 %, 34.81 %, 43.29 %, 16.99 %, and 16.99 % for MSE, MAE, MBE, RMSE, and NRMSE, respectively. Fig. 13 presents the forecasting results for the four seasons on the DKASC-DG dataset. In particular, the prediction results for a cloudy day in autumn are shown in subplot (b) in Fig. 13. It demonstrates that MSRWKV-2DTCN can effectively capture the trend of output power under changeable weather conditions.

4.7. Robustness verification via feature selection

In this subsection, we assess the robustness of the proposed model by incorporating feature selection using the Pearson correlation coefficient method [52]. The objective is to evaluate how the model's performance is influenced when features with weak correlations to the target variable are excluded, thereby verifying the model's robustness.

Feature selection is a common preprocessing step in machine learning that helps to enhance model performance by eliminating redundant or irrelevant features. By focusing on the most important features, we can reduce the model's complexity, potentially improve predictive accuracy, and prevent overfitting [52,53]. In line with the approach taken by Liu et al. [52] in their work on PV power forecasting, we applied the Pearson correlation coefficient method to identify the most significant features influencing PV power output.

The Pearson correlation coefficient was calculated for each feature in relation to the target variable, Active Power (AP). Fig. 14 shows heatmaps of two datasets. Based on previous research and established practices in the domain [54–56], a threshold of 0.2 was set for the absolute value of the correlation coefficient. Features with an absolute correlation value below 0.2 were deemed to have a negligible linear relationship with the target variable and were therefore excluded from further analysis. As a result, Weather Daily Rainfall (WDR), Wind Direction (WD), and Air Pressure (APre) were excluded from the data used for model training. Removing these features helps reduce potential noise during the training process.

Following the feature selection process, we conducted experiments to compare the performance of MSRWKV-2DTCN using the full set of features versus the reduced set of features (with low-correlation features removed). Table 7 presents a comparison of model performance metrics between the full feature set and the reduced feature set for two datasets, DKASC-DG and DKASC-CA. The MSE and MAE metrics show a very slight improvement when using the reduced feature set for both datasets. For DKASC-DG, MSE decreases from 0.0043 to 0.0042, and MAE decreases from 0.0277 to 0.0275. Similarly, for DKASC-CA, MSE decreases from 0.0047 to 0.0046, and MAE remains constant at 0.0252.

This slight reduction in error suggests that the model's high performance is maintained even with the inclusion of less relevant features, indicating a strong inherent robustness.

4.8. Performance evaluation on other PV power datasets

We performed experiments using publicly available datasets of the Yulara solar system collected from DKASC. To further validate the effectiveness of MSRWKV-2DTCN, we conducted experiments on two additional publicly available datasets from DKASC, which were also used by the DSN [27] and other studies for PV power forecasting. Table 8 shows experimental results, with comparison model results reported in the DSN. The "-" symbol is used to indicate the absence of these specific evaluation metrics in the original references. The MSRWKV-2DTCN achieved the lowest metric values among comparison approaches.

Additionally, to further evaluate the performance of MSRWKV-2DTCN, we present a general comparison in Table 9. These general comparison models used datasets from DKASC for PV power forecasting, including the SVM combined with an intelligent optimization algorithm [51], Extreme Learning Machine (ELM) based models [52], Graph Convolution Network (GCN) [53], CNN based models [54,55], and DSN [27]. Table 9 presents the results and information of comparison methods. In contrast, MSRWKV-2DTCN excels in reducing error rates, highlighting its effectiveness for accurate PV power forecasting.

5. Conclusions

In this study, the MSRWKV-2DTCN is introduced to enhance the accuracy of short-term PV power forecasting. MSRWKV-2DTCN integrates the multi-scale RWKV with a multi-scale 2D TCN to learn both the periodicity and interdependencies of historical data. The proposed model identifies critical periodic patterns through FFT, addresses the limitations of the original RWKV model, and leverages the strengths of TCNs for sequence modeling. This combination enhances the model's ability in learning complex dependencies and improves the accuracy of PV power forecasting, aiding in efficient power grid management. The experimental results consistently indicate that MSRWKV-2DTCN outperforms existing methods in terms of accuracy, making it a promising tool for short-term PV power forecasting.

Future research will focus on integrating PV power forecasting with electricity consumption and load forecasting to develop a comprehensive system for power grid management. This integrated approach can improve the scheduling of power plants and the stability of power grids by providing a more holistic view of power generation and consumption patterns. Additionally, further studies could explore the incorporation of other advanced machine learning techniques and visual data sources to continue improving forecasting accuracy and model robustness.

CRedit authorship contribution statement

Jianhua Hao: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fangai Liu:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Weiwei Zhang:** Writing – review & editing, Writing – original draft, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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